



Developing the CareModel Hub Web Application for Implementing the Kidney Failure Risk Equation (KFRE)

Live CareModel Hub Platform

Open the running demonstration of the CareModel Hub system

[Launch CareModel Hub →](#)

Student Information

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Project Overview

The CareModel Hub (CMH) is a modular clinical decision support platform designed to bridge the gap between validated medical algorithms and their practical use in healthcare. The platform uses a microservice architecture where a Node.js backend manages authentication and user roles while isolated Python services perform clinical calculations such as the Kidney Failure Risk Equation (KFRE). The system follows EU NIS2 cybersecurity principles and supports HL7 FHIR interoperability standards for integration with modern healthcare systems.

System Architecture

CareModel Hub System Architecture

Browser



React Frontend (Bulma UI)



Node.js Backend (Auth + RBAC)



Python Model Service (KFRE)



MongoDB Database

DevOps:

GitHub → CI/CD → Docker → Cloud

Project Poster

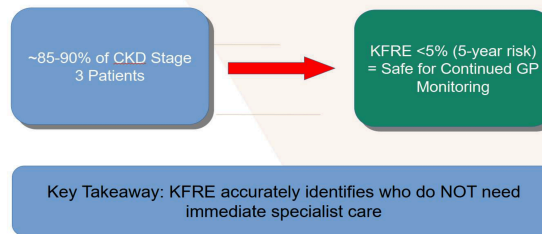
Presenter: Peter Chuk

- **Chronic Kidney Disease (CKD)**, also referred to as **Chronic Renal Disease (CRD)**, affects approximately 12% of the Irish population. Current detection rates remain around 40%, and while only ~3% of CRD patients progress to End Stage Renal Disease (ESRD), the absolute number of patients requiring secondary care services, including in-centre dialysis, continues to rise.
- The majority of CRD patients can be effectively managed within primary care settings, provided that high-risk individuals are identified early.
- The **Kidney Failure Risk Equation (KFRE)** predicts 5-year progression to ESRD using age, sex, eGFR, and ACR. Recommended by NICE NG203, KFRE enhances risk stratification, supports timely intervention, improves resource allocation, and reduces unnecessary secondary referrals.

Regulatory & Governance

- ISO 13485 (QMS) – Defines requirements for a structured Quality Management System for medical device software.
- ISO 14971 (Risk) – Establishes a formal framework for identifying, analysing, and mitigating clinical and technical risks.
- ISO 27001 (InfoSec) – Specifies controls for managing information security and protecting sensitive data.
- MDR / IVDR – EU regulations governing the safety, performance, and clinical validation of medical and in-vitro diagnostic devices.
- GDPR & NIS2 – European regulations ensuring lawful data protection and strengthened Cybersecurity resilience.

KERE: HIGH NEGATIVE PREDICTIVE VALUE



*High negative predictive value (NPV) reduces unnecessary nephrology referrals
Green: Low Risk => GP Monitoring

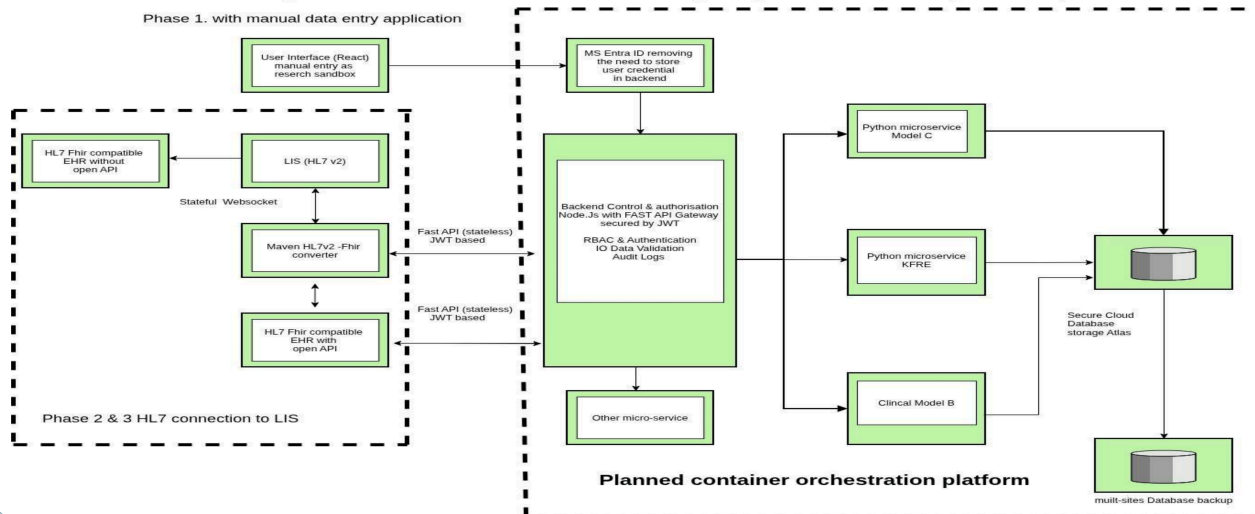
2. System Architecture Overview

Layered microservice architecture:

- React Frontend (UI Layer)
- Node.js (Hapi + TypeScript) Backend
- FastAPI Python Microservice (KFRE Engine)
- MongoDB Atlas Persistence

Design principles: modularity, security, fault isolation, scalability.

A Scalable Design for Clinical Calculations: Future Integrating Manual Data Entry with Hospital Records



CareModel Hub Dashboard New KFRE

Role: user Log out

New KFRE Calculation

MRN: 543210 Specimen No: BC90012
Specimen No must be BC followed by 6 digits

Age: 60 Sex: male eGFR: 35 ACR: 120
eGFR mL/min/1.73m² ACR mg/mmol

Calculate & Save

Scan QR code: <https://pccork.github.io/caremodelhub/>

email	Password	Role
testuser1@caremodelhub.com	TestUserPass123!	user



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Lecturing Team, Department of Computer Science, SETU
Dr. Anita Kealy, project supervisor, SETU

Project Demonstration

peter chuk final project demo, HiDip Computer Science, SETU
Peter Chuk



Watch on

Project Resources

[Download Presentation](#)

[Project Source Code](#)

CareModel Hub - Digital Health Platform for Clinical Risk Models